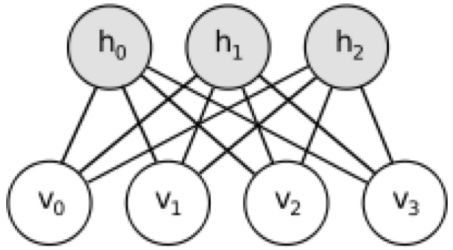
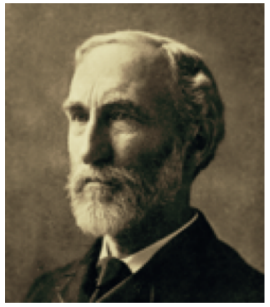
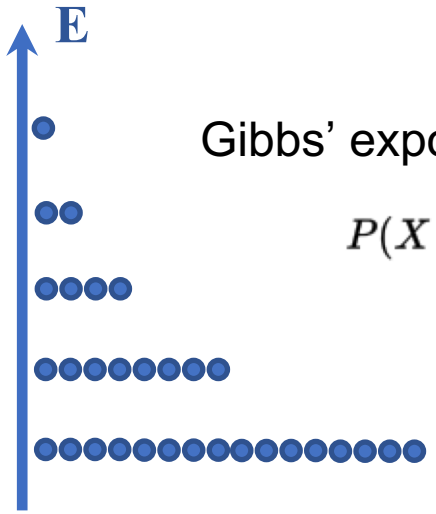
Restricted Boltzmann Machine – two interconnected layers, with restriction that there is no connection inside the layers. **RBM** can learn a probability distribution over its set of inputs.





Josiah Willard Gibbs
1839-1903,New Haven, CT
statistical mechanics

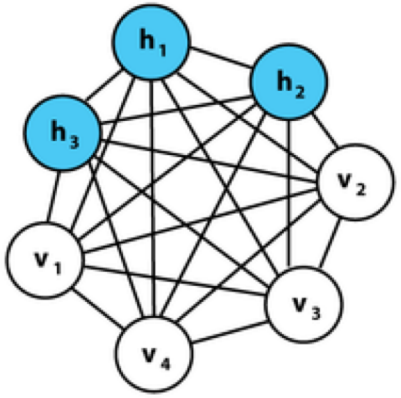


Gibbs' exponential energy distribution

$$P(X = x) = \frac{1}{Z(\beta)} \exp(-\beta E(x))$$

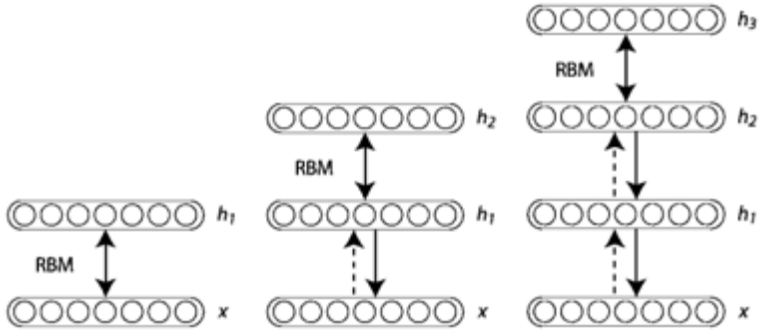
For energy-based models with latent variables **h**, some algorithms are phrased in terms of the negative log - called the **free energy** :

$$\mathcal{F}(\boldsymbol{x}) = -\log \sum_{\boldsymbol{h}} \exp\left(-E(\boldsymbol{x}, \boldsymbol{h})\right)$$



Boltzmann Machine is a fully interconnected network with visible and hidden units, and with an energy defined using Boltzmann distribution.

$$p_i = \frac{e^{-\varepsilon_i/kT}}{\sum_{j=1}^{M} e^{-\varepsilon_j/kT}}$$

Ludwig Boltzmann (1844-1906)
Austrian physicist, developer of statistical mechanics.

Deep Belief Network - training a deep network - one RBM layer at a time